W-L-311a
JANUARY 14, 1959
SUPERSEDING
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May 8, 1956
(See Section 6.)

FEDERAL SPECIFICATION

LIGHT, DESK, INCANDESCENT, FLEXIBLE COLUMN

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers flexible-arm, incandescent, electric desk lights, with weighted base or screw-clamp base.

1.2 Classification.

1.2.1 *Types and classes.*—Desk lights covered by this specification shall be of the following types and classes (see 6.2).

Type I.—Lights, desk, weighted base.

Class 1. — Light, desk, weighted base with parabolic-shaped reflector.
Class 2. — Light, desk, weighted base with trough-shaped reflector.

Type II.—Lights, desk, screw-clamp base.

Class 1.—Light, desk, screw-clamp base with parabolic-shaped reflector.
Class 2.—Light, desk, screw-clamp base with trough-shaped reflector.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

W-L-101 — Lamps, Electric, Incandescent, Large, Tungsten-Filament.
W-L-142—Lampholders (Sockets); Medium-Screw-Shell, General Service.
PPP-B-585—Boxes, Wood, Wirebound.
PPP-B-591—Boxes, Fiberboard, Wood-Cleated.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Boxes, Fiber.
PPP-T-60 — Tape; Pressure Sensitive Adhesive, Waterproof—for Packaging and Sealing.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipments (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver,

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San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–P–116—Preservation, Methods of.

MIL–B–138—Boxes, Wood, Fiberboard-Lined for Overseas Shipment (For Weight of Contents not Exceeding 500 pounds).

MIL–B–10377—Box, Wood, Cleated, Veneer, Paper Overlaid.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

MIL–STD–130 — Identification Marking for U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following publications, of the issues in effect on date of invitation for bids, unless otherwise stated, form a part of this specification:

Underwriters' Laboratories, Inc. Publication:

Standard No. 62—Standard for Flexible Cord and Fixture Wire.

Standard No. 153—Standard for Portable Electric Lamps.

(Copies may be obtained upon application to Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois; 161 Sixth Avenue, New York, N. Y.; 1655 Scott Lane, Santa Clara, California.)

3. REQUIREMENTS

3.1 Preproduction sample.—Unless otherwise specified in the invitation to bid, before production is commenced a sample of the finished commodity shall be submitted or made ready for approval to the contracting officer or his authorized representative. When the initial sample is approved, it shall be used as a control guide throughout production of the contract. This sample may be included in the quantity required for the shipment.

3.2 Fire and casualty hazards.—The bidder shall submit to the contracting agency proof that the desk lights he proposes to supply under this specification conform to Underwriters' Laboratories, Inc., Standard No. 153, as regards to fire and casualty hazards. The label or listing of the Underwriters' Laboratories, Inc., will be accepted as evidence that the desk lights comply with this requirement. In lieu of the label or listing, the bidder may submit independent proof satisfactory to the contracting agency that desk lights conform to the published standards, including methods of test of the Underwriters' Laboratories, Inc. Compliance with the above requirements as regards to fire and casualty hazards, does not absolve the bidder from complete compliance with other requirements of this specification, in order to secure acceptance of desk lights.

3.3 Design.

3.3.1 *General.*—The type I light shall consist of a weighted base and an attached column at the upper end to which there shall be a flexible tubing arm supporting a lampholder and reflector. The type II light shall conform to the requirements of type I except it shall have a clamp-type base.

3.3.2 *Current.*—Lights shall be furnished for operation with 110 – 125 volt alternating current.

3.3.3 *Size.*—The lights shall be not less than 24 inches nor more than 28 inches high when measured from the bottom of the base along the axis of the column to the furthest point on the rim of the reflector. Flexible tubing of

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the column shall not be less than 9 inches in length.

3.4 **Material.**—Lights shall be constructed of metal and shall be complete with all necessary wiring and auxiliaries but without the lamps. Bases, columns and supports shall be of one or more of the following: Iron, steel, brass, bronze, aluminum, aluminum alloy, zinc alloy, or magnesium alloy. White-metal casting (lead base) shall not be used in bases, columns, or supports. Materials not definitely specified shall be of a good quality normally used by manufacturers for similar applications. Materials shall be new and free from defects or imperfections which might affect the serviceability of the finished unit.

3.5 **Construction.**—Each light shall consist of a metal base with an upright metal column or other support, on the upper end of which there shall be mounted a reflector and reflector housing; the reflector may be integral with the housing or may be separate and enclosed by a housing. The light shall be sufficiently sturdy to withstand without damage the application of any force or stress to which it will be normally subjected in service and shipment.

3.5.1 *Exposed edges.* — Exposed edges of sheet metal or other metal parts shall be so constructed that personal injury due to sharp edges will not occur.

3.5.2 *Wiring.*—Wiring other than the attached cord and plug shall be entirely concealed. Attached cord shall enter the light through or at the base.

3.5.3 *Bases.*

3.5.3.1 *Base, type I.*—The base shall be of cast, pressed or spun metal and the minimum wall thickness shall be as follows:

	Inch
Gravity cast metal base ----	0.156
Pressure cast metal base ---	0.065
Pressed or spun metal base -	0.0359

The base area and weight shall be sufficient to cause the light to be free standing on a horizontal surface and to prevent the light from overturning when inclined at an angle of 8 degrees from the horizontal in any direction as required by Underwriters' Laboratories, Inc. The bottom of the base shall be a plane surface and shall be covered with a dark-colored felt at least one-sixteenth of an inch thick or be provided with other suitable means to prevent scratching of the surface on which the light rests.

3.5.3.1.1 *Weighted filling.* — Concrete or cast-metal filling shall be securely attached to the inside of the base to produce a light which shall be free standing under the conditions stated in 3.5.3.1.

3.5.3.2 *Base, type II.* — Base shall be of formed steel or cast iron and shall be equipped with steel mounting screws which will permit opening up to at least 2 inches. Felt specified in 3.5.3.1 shall be permanently affixed to the two opposing binding surfaces of the clamp.

3.5.4 *Column.*—The light column shall be of metal and shall consist of an upright attached to the base and a flexible arm attached to the upper end of the upright by a threaded adaptor, or it shall consist of a flexible arm attached to the base. Reflector housing shall be attached at upper end of column.

3.5.4.1 *Movement of column.*—The column shall permit the reflector housing to be moved to direct the rays of the light through a vertical angle of not less than 90 degrees to include vertically downward and horizontally. The reflector housing shall not be rotated with respect to the column axis to accomplish this. The column shall remain in any normally selected position.

3.5.5 *Reflector housing and reflector.*—The reflector housing and reflector shall be of metal and shall be securely fastened to the flexible tubing or the lampholder at its upper

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end by clamp or snap-on method. The reflector housing and reflector may be integral or consist of a separate housing and reflector. The size and construction shall permit the insertion and removal of a standard 60 watt medium screw base incandescent lamp conforming with Federal Specification W–L–101. When inserted into the socket, no portion of the lamp shall project beyond the plane of the reflector rim.

3.5.5.1 *Movement of reflector housing.* — Reflectors shall be attached to the upper end of the extension arm by means of a socket to permit easy rotation of the reflector in all directions and to hold the reflector firmly in all positions.

3.5.5.2 *Class 1, parabolic-shaped reflector.* —The reflector shall be parabolic or bowl shaped and the plane bounded by the rim shall be essentially circular.

3.5.5.3 *Class 2, trough-shaped reflector.*— The trough shaped reflector shall have an overall length and depth to accommodate the size lamp specified in 3.4.5 in a horizontal position inside the reflector. The rim of the reflector shall be on a single plane.

3.5.6 *Electrical components.*

3.5.6.1 *Lampholder.*—The lampholder shall conform with the requirements of type I, class A or C of Federal Specification W–L–142, except the shell may be of brass or aluminum.

3.5.6.2 *Attachment-plug-cap and cord.*—A cord having an exposed length of not less than 7 feet with a molded-on type, rubber, neoprene or thermoplastic insulated, parallel blade attachment-plug-cap, shall be provided with the desk light. The cord shall be N. C. C. type SP–1, SPT–1, SP–2 or SPT–2, integral construction, 18 AWG, as provided for in Underwriters' Laboratories, Inc., Standard No. 62.

3.5.7 *Finish.*—All exposed metal parts shall be cleaned and free of dirt, oil, grease, and moisture before applying finish. All surfaces, except aluminum reflecting surfaces, shall be covered completely with one or more coats of enamel or by electro plating. Reflecting surfaces other than aluminum shall be white and all other enameled surfaces shall be brown crackle, bronze or gun-metal finish, as specified. Enameled surfaces shall be free from tears, star marks, crazing, blisters, hairlines, chipping, pinholes and other irregularities or defects.

3.5.8 *Marking.* — The lights shall be marked in accordance with Military Standard MIL–STD–130.

3.6 Workmanship.—Lights shall be free from defects which detract from appearance or impair serviceability. The entire reflector and light assembly shall be firmly seated at all points and locked with nuts and washers wherever necessary.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 General.—The quality assurance provisions contained in this section are applicable to inspection performed by or for the Government. Quality assurance provisions for components or materials described shall be in accordance with this specification and with subsidiary specifications, figures, and standards referenced herein to the extent applicable, except that this specification shall govern in case of conflict.

4.2 Preacceptance inspection.

4.2.1 *Preproduction inspection.*—When required, inspection and testing of the preproduction sample shall be made of a completely fabricated desk light, for all provisions of this specification applicable to end product examination and test.

4.3 Inspection for acceptance.—Inspection will be made in accordance with the provi-

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sions set forth in Military Standard MIL–STD–105, except where otherwise indicated hereinafter.

4.3.1 *Inspection of components.*

4.3.1.1 *Visual examination.*—Each component or material which cannot be checked after assembly shall be examined in lots or batches, by sampling, for compliance with this specification and with applicable specifications and figures. Any deviation therefrom shall constitute a defect. The acceptance quality level shall be 4.0 defects per 100 units. The inspection level (sample size) applicable to each lot or batch, shall be level II from table III of Military Standard MIL–STD–105.

4.3.1.2 *Dimensional examination.* — Each component or material which cannot be measured after assembly shall be examined in lots or batches by sampling for compliance with this specification and with applicable specifications and figures. Any deviation therefrom shall constitute a defect. The acceptance quality levels shall be 4.0 defects per 100 units. The inspection level (sample size) applicable to each lot or batch, shall be level L–3 of the Appendix to Military Standard MIL–STD–105.

4.3.1.3 *Flexibility examination.*—With the flexible column bent so that the light is at the maximum horizontal distance from the center line of the junction of the flexible column and the base, there shall be no tipping or instability. The unit shall be capable of passing the aforementioned test for any radial position of the flexible column. This determination shall be made on the preproduction sample or an early production unit prior to acceptance of the first inspection lot and need not be repeated during the life of the contract unless there is a change in design, construction or workmanship.

4.3.2 *End item inspection.*

4.3.2.1 *Visual examination.* — The defects found during examination shall be classified in accordance with the list shown in 4.3.2.1.1. The sample unit for this examination shall be one light.

4.3.2.1.1 *Examination for visual characteristics.*

Examine	Defects	Classification	
		Major	Minor
Finish	Not cleaned before applying finish	X	
	Surfaces other than aluminum not coated with enamel or electro-plated	X	
	Reflecting surfaces other than aluminum not white enamel coated	X	
	Type of finish not as specified or blistered, peeled, chipped or area of no film or thin film		X
	Stained, overspray, wet or tacky or foreign matter imbedded in finish		X
	Area of rust		X
Material	Any component not fabricated of the specified material	X	
Design	Any characteristics not in accordance with specified requirements		X
Construction and workmanship	Any part missing or not specified type or size	X	

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Examine	Defects	Classification	
		Major	Minor
General (applicable to all components and assemblies)	Fractured, split, malformed castings, or otherwise impaired seriously affecting serviceability	X	
	Defective or otherwise damaged but not seriously affecting serviceability		X
	Any part misplaced, loose or not in proper alignment		X
	Any sharp burrs, sliver or edge that may cause injury to personnel	X	
	Any operation omitted or not properly performed, seriously affecting serviceability	X	
	Any operation omitted or not properly performed, not seriously affecting serviceability		X
	Any functioning component that is inoperative or will not operate as intended	X	
	Not connected or joined as specified, or poorly accomplished, not seriously affecting serviceability	X	
Workmanship of electrical equipment	Equipment does not comply with applicable UL standard	X	
	Electrical connection not securely attached	X	
	Electrical cord damaged (insulation or wire worn, cracked, cut, burned or broken)	X	
	Light tilts over when flexible arm is placed in horizontal position (i.e., base weight not heavy enough to support light	X	
	Flexible tubing does not support weight of attached reflector and standard 60 watt bulb in any fixed position	X	
	Any portion of a 60 watt bulb projecting beyond plane of the reflector rim		X
Castings	Crack or other defect seriously affecting serviceability	X	
	Defective but not seriously affecting serviceability. Slight malformation in non-critical area that will not reduce strength		X
Bolts, nuts, screws and other types of threaded components	Missing, not specified type, broken, stripped, fractured or loose, seriously affecting serviceability	X	
	Lock nut or washer missing where required		X
Marking	Missing, incomplete, not legible, not specified type or size, not in proper location, not affixed in the prescribed manner, not correct color, plate, label, tag, etc., not fabricated of specified material, not stenciled, stamped, embossed, printed, lithographed, whichever is required		X

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4.3.2.2 *Dimensional examination.*—Any dimension that is not within the specified tolerance shall be classified a defect.

4.3.2.3 *Inspection levels and AQL's.*—Inspection levels and AQL's (expressed in defects per 100 units) shall be as follows:

	AQL	Inspection level
For examination in 4.3.2.1—Major	2.5	II
For examination in 4.3.2.1—Totals	6.5	II
For examination in 4.3.2.2—Class	4.0	L-3

4.3.2.4 *Functional determination.*—Testing shall be performed on sample desk lights, equipped with lamps, to determine compliance with intended functions. Any malfunction of the sample desk light shall be classified as a defect. The inspection level (sample size) applicable to each end item inspection lot shall be L-4 of the Appendix to Military Standard MIL-STD-105. The acceptable quality level shall be 4.0 defects per 100 units.

4.4 **Inspection and tests of preparation for delivery requirements.** — The preservation, packaging, packing and marking of the equipment shall be subject to inspection and tests by Government inspectors to determine compliance with the requirements of section 5 of this specification. The contractor shall provide the necessary facilities and supplies for inspection and tests made prior to final acceptance.

5. PREPARATION FOR DELIVERY

(Nonmilitary agencies.—Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment).

5.1 Packaging.

5.1.1 *Level A.*—Each light shall be packaged in accordance with Military Specification MIL-P-116, method IC-5, utilizing containers conforming to class 2, W5c of Federal Specification PPP-B-636. The containers shall be closed with adhesive and stitching as specified in the appendix to the box specification and in addition, all corners and joints shall be waterproof sealed with 2-inch minimum width tape conforming to type II, class 1 of Federal Specification PPP-T-60.

5.1.2 *Level C.*—Lights shall be packaged in accordance with the manufacturer's normal practice.

5.2 Packing.

5.2.1 *Level A.*—Twelve packages (12 lights of one type and class only, packaged as specified in 5.1) shall be packed in the most compact manner in a snug-fitting shipping container constructed and strapped to conform to class 2, V3c or V3s of Federal Specification PPP-B-636. Alternatively, shipping containers constructed, closed, and strapped to conform to overseas type of Federal Specifications PPP-B-591, PPP-B-601, or Military Specification MIL-B-10377; or class 2, style 2 or 4 of Federal Specification PPP-B-621; or style 1, 2, or 2A, class 3 use of Federal Specification PPP-B-585; or Military Specification MIL-B-138 may be used. Only one type, style, and size shipping container shall be used for any one item on any one contract.

5.2.2 *Level B.*—Twelve packages (12 lights of one type and class only, packaged as specified in 5.1) shall be packed as specified in 5.2.1, except that the shipping container shall conform to type I or II, class 1, style RSC of Federal Specification PPP-B-636 except that the minimum bursting strength of the fiberboard shall be 200 p.s.i. All flaps of the fiberboard shipping container shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps, or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive. Alternatively, shipping containers constructed to conform to domestic type style A or B of Federal Specification PPP-B-591, PPP-B-601, or Military Specification MIL-B-10377; or class 1, style 2 or 4 of Federal Specification PPP-B-621;

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or style 1, 2, or 2A, class 1 use of Federal Specification PPP–B–585 may be used.

5.3.3 *Level C.* — Packages which require overpacking for acceptance by the carrier shall be packed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to point of delivery called for in the contract or purchase order.

5.3.1 *Military agencies.*—Interior packages and shipping containers shall be marked in accordance with the requirements of Military Standard MIL–STD–129.

5.3.2 *Nonmilitary agencies.*—Marking for shipment shall be in accordance with Federal Standard No. 123.

6. NOTES

6.1 **Intended use.**—The lights are intended for use on desks, tables, benches, machines, et cetera to shield the light from the worker's eyes and direct maximum intensity on the work area.

6.2 **Ordering data.**—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number and date of this specification.

(*b*) Stock number.

(*c*) Type and class of light required (see 1.2).

(*d*) If a preproduction sample is not desired, the invitation should so indicate (see 5.1 and 5.2).

(*e*) Both the level of packaging and the level of packing required, (see 5.1 and 5.2).

6.3 **Transportation description.**—Transportation description applicable to this item is:

Lamps, electric, incandescent.

Carload minimum weights 15,000 and 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor volume minimum weight, any quantity.

6.4 This specification incorporates the requirements of Military Specification MIL–L–1989, Lamp, Desk, Flexible Arm (Army), dated 31 January 1950.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
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Air Force.

Copies of this specification may be purchased for 5 cents each.

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